Assignment 2:

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#include <iostream>
#include <vector>
#include <cmath>
using namespace std;

// Check if a number is a power of 2
bool isPowerOfTwo(int x)
{
    return (x && !(x & (x - 1)));
}

int main()
{
    string data;
    cout << "Enter data bits: ";
    cin >> data;

    int m = data.length();
    int r = 0;

    // Calculate required parity bits
    while (pow(2, r) < (m + r + 1))
        r++;

    int totalBits = m + r;
    vector<int> hamming(totalBits + 1);

    // Insert data bits and leave parity bit positions empty
    int j = 0;
    for (int i = 1; i <= totalBits; i++)
    {
        if (isPowerOfTwo(i))
            hamming[i] = 0;
        else
            hamming[i] = data[m - 1 - j++] - '0';
    }

    // Calculate parity bits (Even Parity)
    for (int i = 0; i < r; i++)
    {
        int p = pow(2, i);
```

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    int parity = 0;

    for (int k = 1; k <= totalBits; k++)
    {
        if (k & p)
            parity ^= hamming[k];
    }

    hamming[p] = parity;
}

cout << "\n----- TRANSMITTER -----\n";
cout << "Generated Hamming Code: ";

for (int i = totalBits; i >= 1; i--)
    cout << hamming[i];

cout << endl;

// Receiver
string received;
cout << "\nEnter received codeword: ";
cin >> received;

vector<int> recv(totalBits + 1);

for (int i = totalBits, idx = 0; i >= 1; i--, idx++)
    recv[i] = received[idx] - '0';

int errorPos = 0;

// Check parity
for (int i = 0; i < r; i++)
{
    int p = pow(2, i);
    int parity = 0;

    for (int k = 1; k <= totalBits; k++)
    {
        if (k & p)
            parity ^= recv[k];
    }

    if (parity)
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            errorPos += p;
        }

        cout << "\n----- RECEIVER -----\n";

        if (errorPos == 0)
        {
            cout << "No error detected." << endl;
        }
        else
        {
            cout << "Error detected at bit position: " << errorPos << endl;

            recv[errorPos] ^= 1;

            cout << "Corrected Codeword: ";
            for (int i = totalBits; i >= 1; i--)
                cout << recv[i];
            cout << endl;
        }

        cout << "Recovered Data Bits: ";

        for (int i = totalBits; i >= 1; i--)
        {
            if (!isPowerOfTwo(i))
                cout << recv[i];
        }

        cout << endl;

        return 0;
    }
```